CSE 5106 Proposal - Preventing Manipulation of Prediction Markets

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1 Introduction

Prediction markets are financial venues designed to aggregate dispersed information regarding future events. Over the past decades, these markets have drastically grown in popularity, with examples such as Kalshi and PolyMarket raising multi-billion dollar valuations. Prediction markets allow traders to bet on the outcome of future events in areas as diverse as sports, economics, and politics. Prediction markets trade binary options that pay a fixed amount (typically \$1.00) if some event occurs and \$0.00 otherwise [1]. As such, the current price of an option corresponds to the estimated probability of that event occurring. Traders holding shares can choose to sell the share at the current price or wait until the event has concluded to get a flat \$0.00 or \$1.00. With free-market trading, these prices and probabilities represent the real-time market consensus on the projected outcome of an event.

Our project aims to simulate prediction markets with an emphasis on how markets can be made robust to attempts at manipulation by an adversarial agent. Trades in the market will be performed by agents with various attributes that will determine how they respond to changes in both price and market probability. The adversarial agent will attempt to manipulate the market by buying and selling shares in bulk to elicit an over-response from other agents in the market. This seeks to cause a significant deviation from the adversarial agent's estimation of the true probability of the event, allowing the adversary to profit by selling the inflated shares before the market returns to equilibrium. Such market manipulation may negatively impact the perceived repuation of the prediction market, and it is therefore in the interest of the market organizers to prevent such attacks.

Existing research has explored the susceptibility of prediction markets to manipulation. Notably, [2] provides a game-theoretic model where agents can collude to make market prices uninformative, specifically in cases where agents can influence the event's actual outcome. Furthermore, [3] provides experimental evidence indicating that well-funded manipulators can effectively destroy a prediction market's ability to aggregate information purely through financial pressure. While these studies confirm that manipulation is a fundamental threat to market accuracy, literature focusing on automated mechanism design for active defense remains limited.

2 Approach

Trades in a prediction market are aggregated by an Automated Market Maker (AMM) to update the price of the option. An example cost-function-based AMM is the Logarithmic Market Scoring Rule (LMSR) [5]. LMSR and other AMMs are used to provide and control liquidity at a given price.

We represent the current state of the market as $q = (q_1, q_2, \dots, q_n)$ where each q_i represents the number of outstanding shares for outcome i. When an order is placed, the AMM mints the outstanding shares for that order; if the position is exited, the AMM burns those shares. The AMM prices the shares according to some convex cost function:

$$p_i(q) = \frac{\partial C(q)}{\partial q_i}$$

For example, an implementation of LMSR used by the Gensyn AI team in a market designed for algorithmic predictions is [4].

$$C(q) = b \ln \left(\sum e^{q_j/b} \right), \quad p_i = \frac{e^{q_i/b}}{\sum e^{q_j/b}}$$

We consider a prediction market with two types of agents. The first group of agents place trades according to the current option price p_i and its rate of change $p^t - p^{t-1}$ and noisy access to some external signal $v(t)$ representing the true probability of the event occuring. The sensitivity to these influences may vary between agents of this type. Agents in this group have various degrees of rationality: some may buy simply because the price is rising quickly. The second type of agent is an adversarial manipulator who wishes to force the market to represent a false probability. This may be done to temporarily force the market into a state significantly deviating from $v(t)$ such that the manipulator can profit when the market corrects itself to $v(t)$. Formally, the adversary's goal G is to set outcome k to be above a threshold τ.

$$G = \{q \in S \mid p_k(q) \geq \tau\}$$

The attacker can achieve this by purchasing a bundle of options Δq, which influences p_i by affecting the number of outstanding shares. The cost of this action is given by the money deposited into the AMM. For a binary event (i.e., $q = (q_1, q_2)$) with no additional constraints on options purchased, the attacker's problem simply reduces to how many shares must be bought to update the price to some threshold. However, for an event with a greater number of mutually exclusive outcome (e.g., which candidate will win a primary), the attacker can choose to raise the price of option i either by buying it directly or by selling shares of option $j \neq i$. If the liquidity distribution across options is not uniform, the attacker's problem is finding the optimal path π of buying and selling shares to achieve G.

$$\min_{\pi} \mathrm{Cost}(\pi, b)$$

Where the cost function includes the price of buying shares in addition to additional fees or constraints Φ imposed by the AMM:

$$\mathrm{Cost}(\pi, b) = \sum_{(q,q') \in \pi} \left(C(q') - C(q) + \Phi(q, q', b) \right)$$

To prevent this attack from occurring, the AMM is able to modify the liquidity parameter b. However, the AMM is subject to higher loss for increasing b and the market may fail to represent a realistic estimate of the probability of some event. Furthermore, the AMM must distinguish between manipulation attempts and legitimate sudden shifts in news. Failure to do so would result in an inaccurate market. This is represented as $\mathrm{Risk}(b)$ and is weighted by the AMM's risk-aversion parameter λ. Thus, the problem of the AMM (defender) becomes

$$\max_{b} \left(\min_{\pi} \mathrm{Cost}(\pi, b) - \lambda \, \mathrm{Risk}(b) \right)$$

If the minimum cost of executing the manipulation is above the budget of the attacker, the attack is interdicted. The total space of all q may be very large and computationally intractable, so a constraint generation approach such as in [6] or other related techniques will be required.

3 Goals and Outcomes

The primary objective of this project is to create a prediction market simulation where autonomous agents trade binary outcome contracts based on their beliefs about future events. The simulation will implement:

1. Prediction Market Environment: We will implement LMSR as a baseline AMM, though we leave the option to explore other AMM mechanisms open. An AMM ensures that even in low-volume scenarios, the market provides continuous liquidity and dynamically adjusts prices (p_i) based on the cost function $C(q)$.

2. Autonomous Agents: Our agents will use the Kelly Criterion to determine optimal position sizing given their noisy access to the true probability $v(t)$, current market prices, and risk tolerance. This simulates realistic wealth management, allowing the project to act as a study of market evolution.
3. Adversarial Agents: We will model manipulators attempting to force the market away from an equilibrium to make a profit. The simulation will test the AMM's ability to interdict these attacks by dynamically adjusting the liquidity parameter b. We will measure how price accuracy degrades as the adversary's budget increases relative to the AMM's risk-aversion parameter (λ).

Our project aims to provide insights on how Prediction Markets can be made resilient to manipulation. The simulation will reveal the optimal liquidity depth (b) and fee structures required to make market manipulation economically non-viable while maintaining enough volume for price discovery.

4 Timeline

The project will be executed over a 12-week period. We will spend the first three weeks implementing a robust environment and AMM and finalizing the adversary's manipulation strategy. This will include implementing the LMSR cost function and state space $q = (q_1, \ldots, q_n)$ and developing the core trading platform to handle minting/burning of shares and price updates (p_i). We will ensure that market prices respond accurately to manual trades before programming autonomous agents with Kelly Criterion logic for position sizing. We will also establish the true signal $v(t)$ and assign various degrees of access to this signal to different agents groups.

We expect to schedule the core of the project over six to eight weeks. This will involve the introduction of adversarial agents who attempt to manipulate the market and implement the AMM defender logic to dynamically adjust the liquidity parameter and collect data for the impact on manipulation cost. We will allocate more time for this phase as it is the core of the content of our final report.

In the remaining time, we will wrap up the project by analyzing the sensitivity of the AMM's risk-aversion parameter and finalize the project report, and summarize all collected data into clear visuals.

References

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